Omin Kwon

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Research Interest

Building efficient and scalable ML systems through algorithm-system co-design.

Education

Seoul National University Sept 2025 – present

M.S. in Computer Science and Engineering

- Advisor: Prof. Jae W. Lee, ARC Lab
- GPA: 4.3/4.3

Seoul National University Mar 2019 – Aug 2025

B.S. in Electrical and Computer Engineering (summa cum laude)

- GPA: 4.26/4.3 (upper-division), 3.95/4.3 (overall)
- Ranked 1st in Introduction to ML, Computer Organization & Design, and Computer Networks.
- Leave of absence for service in the Republic of Korea Air Force (May 2021-Feb 2023)

Hansung Science High School Mar 2017 – Mar 2019

Early Graduation for Academic Excellence

Publications

SketchSSM: Write to the Full State, Read from a Compact Sketch
Omin Kwon, JoongWon Shin, Minseo Kim, Kurt Keutzer, Sehoon Kim†, Jae W. Lee†
NeurIPS 2026 LCFM Workshop (**Oral**); Under Review
Project Page · † Co-corresponding authors.

HERALD: High Throughput Block Diffusion LLM Inference via CPU-GPU Cooperative KV Cache Retrieval
Omin Kwon, Doyeon Kim, Jongseok Park, Seung Yul Lee, Sehoon Kim, Ion Stoica, Jae W. Lee
Under Review
Paper

Triage: Skip, Skim, or Refine Attention Tiles in Video Diffusion Transformers
Yeonjae Kim, **Omin Kwon**, Seong Hoon Seo, Jae W. Lee
Under Review
Paper

MAGE: All-[MASK] Block Already Knows Where to Look in Block Diffusion LLM
Omin Kwon, Yeonjae Kim, Doyeon Kim, Minseo Kim, Yeonhong Park, Jae W. Lee
NeurIPS 2026; ICML 2026 AdaptFM Workshop (**Oral**, **Best Paper Award (1/218 = 0.46%)**)
Paper

STAR: Improving Lifetime and Performance of High-Capacity Modern SSDs Using State-Aware Randomizer
Omin Kwon*, Kyungjun Oh*, Jaeyong Lee, Myungsuk Kim, Jihong Kim
ICCAD 2025 (**Oral**)
Paper · * Equal contribution. U.S. patent application with SK hynix: No. 19/569899, filed March 17, 2026.

NestedFP: High-Performance, Memory-Efficient Dual-Precision Floating Point Support for LLMs
Haeun Lee, **Omin Kwon**, Yeonhong Park, Jae W. Lee
NeurIPS 2025
Paper

AiDE: Attention-FFN Disaggregated Execution for Cost-Effective LLM Decoding on CXL-PNM
KyungSoo Kim, **Omin Kwon**, Yeonhong Park, Jae W. Lee
IEEE CAL
Paper

Research Experience

Graduate Student Researcher Sept 2025 – present
Seoul National University Seoul, South Korea

- Research visit to FAST Lab, UIUC (Mar-May 2026).
- **MAGE (Project Lead)** · Advisor: Prof. Jae W. Lee
  - Designed a training-free **sparse attention algorithm** for block diffusion LLMs.
  - Developed an **inference system** integrated into SGLang with optimized CUDA kernels.
  - Achieved up to 6.82× end-to-end speedup at 128K context with near-lossless accuracy.
- **HERALD (Project Lead)** · Advisor: Prof. Jae W. Lee
  - Designed a training-free **sparse attention algorithm** for block diffusion LLMs with KV offload.
  - Developed an **inference system** integrated into SGLang with a dual-stream CPU-GPU pipeline.
  - Achieved up to 2.28× decode throughput over GPU-only serving with near-lossless accuracy.
- **SketchSSM (Project Lead)** · Advisors: Prof. Jae W. Lee and Prof. Sehoon Kim
  - Designed a training-free **linear attention algorithm** that reduces state-read I/O.
  - Developed an **inference system** integrated into vLLM with optimized CUDA kernels.
  - Achieved up to 2.77× decode throughput over Standard vLLM with near-lossless accuracy.

Undergraduate Student Researcher Sept 2024 – Aug 2025
Seoul National University Seoul, South Korea

- **NestedFP** · Advisor: Prof. Jae W. Lee
  - Integrated the NestedFP CUDA kernel into vLLM with a workload-aware precision-switching scheduler.
- **AiDE** · Advisor: Prof. Jae W. Lee
  - Proposed workload-aware KV cache placement for CXL-PNM and developed an LLM simulator.
- **STAR (Project Lead)** · Advisor: Prof. Jihong Kim
  - Proposed a 3D NAND randomization algorithm that extends SSD lifetime by up to 2.3×.
  - Designed STAR hardware and integrated it into the SSD controller.
  - Key contributor to a U.S. patent application with SK hynix on STAR.

Teaching Experience

Teaching Assistant, Advanced Computer Architecture (4190.571) Sept 2026 – present
Seoul National University Seoul, South Korea

Undergraduate Course Tutor Mar 2024 – June 2025
Seoul National University Seoul, South Korea

- Selected as an official departmental tutor based on academic excellence.
- Courses: Computer Organization and Design, Digital System Design, and Microelectronics.

Honors & Awards

Best Paper Award @ ICML26 AdaptFM Workshop July 2026
ICML 2026 AdaptFM Workshop, Best Paper among 123 accepted papers from 218 submissions (0.46%).

Grand Prize (Minister of Science and ICT Award), 2024 AI Semiconductor Idea and Design Contest Jan 2025
Ministry of Science and ICT, designed a resolution-aware diffusion accelerator. News: Edaily.

Grand Prize (Deputy Prime Minister and Minister of Education Award), 2024 STOB League Semiconductor Contest Dec 2024
Ministry of Education, designed a state-aware data randomizer for high-performance SSDs. News: Dong-A Ilbo.

Grand Prize, 2024 FriendliAI LLM Hackathon July 2024
FriendliAI, developed a knowledge-graph RAG application using the FriendliAI API.

SNU Specialized Semiconductor College Scholarship Aug 2024 – Aug 2025
Seoul National University

Merit-Based Scholarship 2023, 2024
Seoul National University